DM – Tree

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Outline

1. Definitions and examples
   - Tree
   - Rooted tree
   - Tree model examples
2. Properties

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Tree

Definition

A tree *is a* connected *simple graph with no circuits.*

Examples:

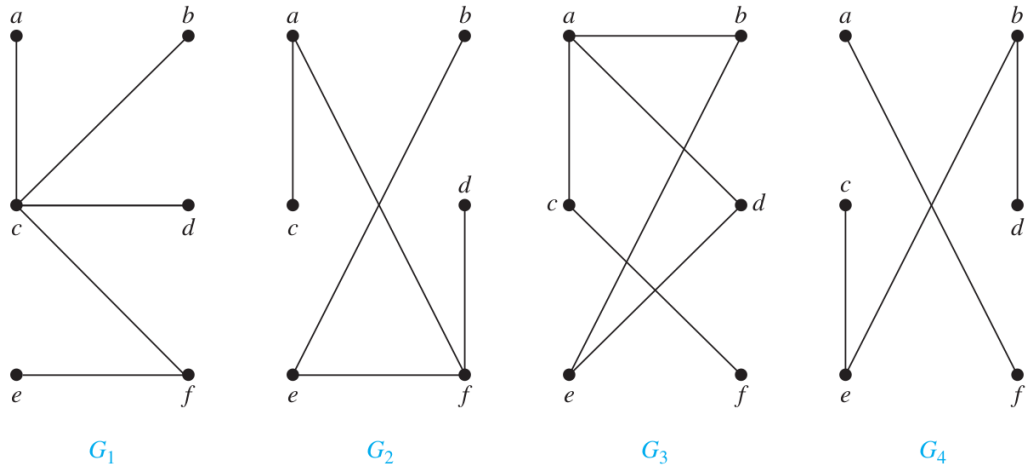


Figure: Rosen, p. 746

- The graphs G_1 and G_2 are trees.
- The graph G_3 is not a tree (*abed* is a circuit).
- The graph G_4 is not a tree (not connected).

Tree

Theorem

A simple graph is a tree if and only if there is a unique simple path between any pair of its vertices.

Examples:

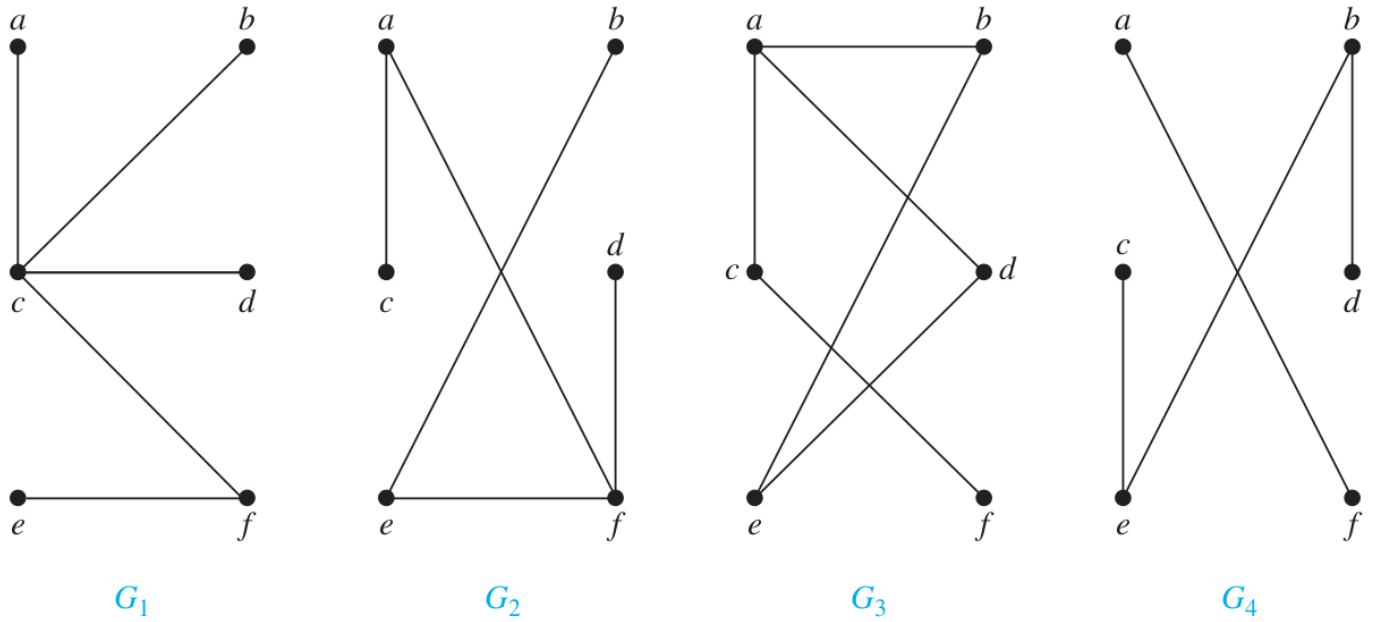


Figure: Rosen, p. 746

Forest

A graph which contains no circuits, but which is not necessarily connected, is called a *forest*. Each connected component of a forest is a tree.

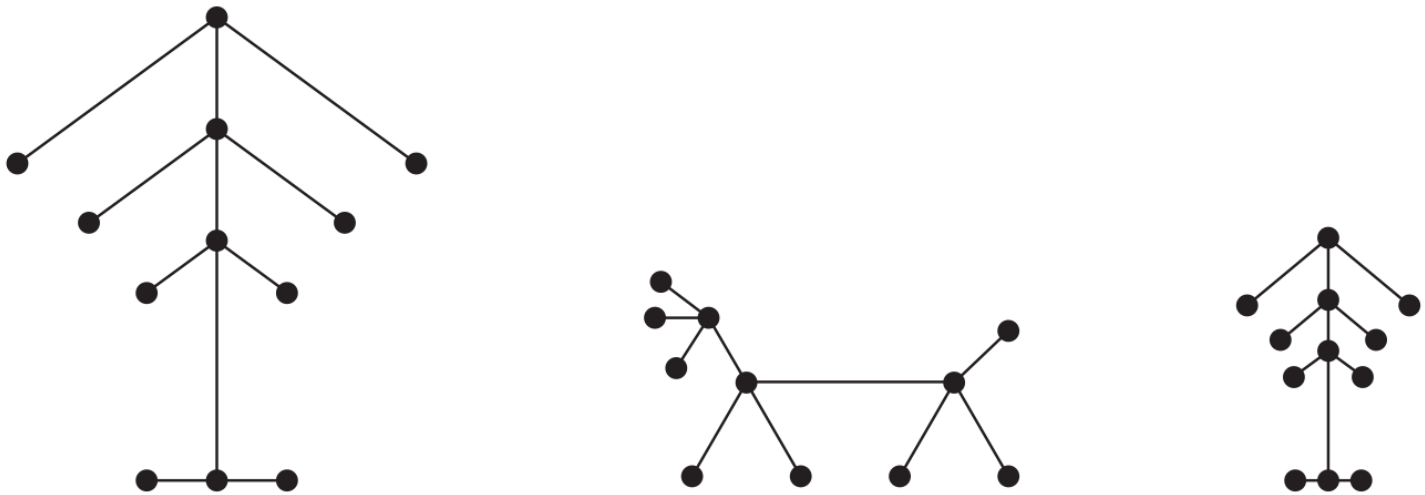


Figure: Rosen, p. 746

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Rooted tree

Definition

A rooted tree is a tree in which one vertex has been designated as the root and the edges are directed away from the root.

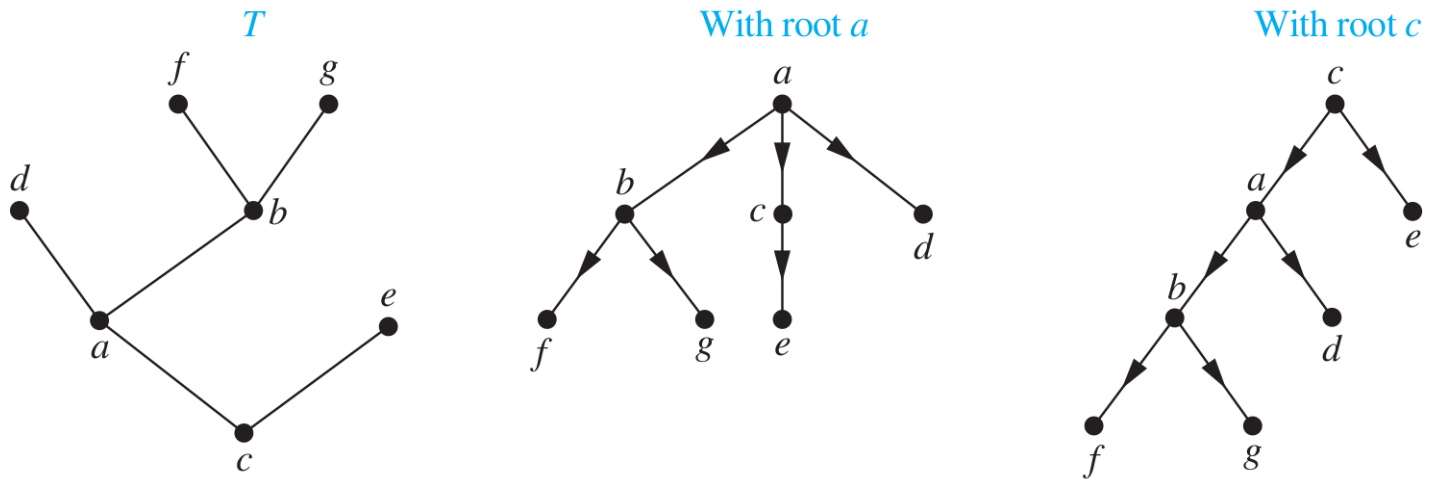


Figure: Rosen, p. 746

Rooted tree

Definition

If (u, v) is an edge in a rooted tree, then u is the parent of v, and v is a child of u.

The vertices with no children are the leaves. The vertices which have children are the internal vertices.

The ancestors of a vertex u are the vertices on the path from the root to u. The decendants of u are the vertices having u as an ancestor.

The subtree rooted at u is the subgraph induced by u and its decendants.

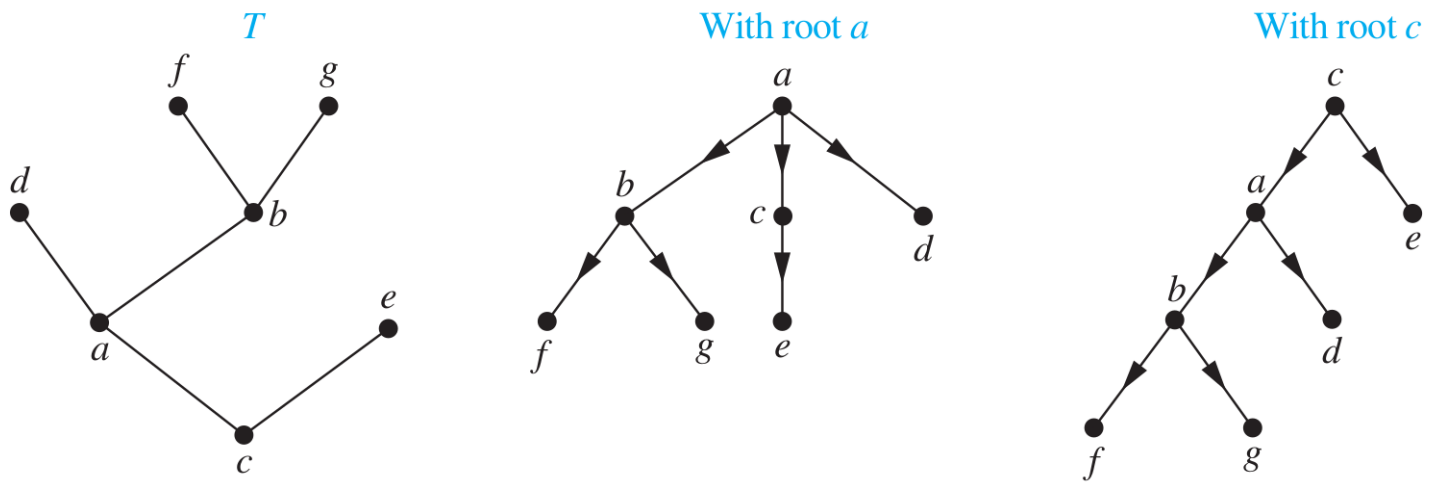


Figure: Rosen, p. 746

m-ary tree

Definition

A rooted tree is a m-ary tree if every vertex has at most m children.

An m-ary tree is full if every internal vertex has exactly m children.

An m-ary tree with $m = 2$ is called a binary tree.

Remark: If $m < n$, then an m-ary tree is also an n-ary tree.

Examples:

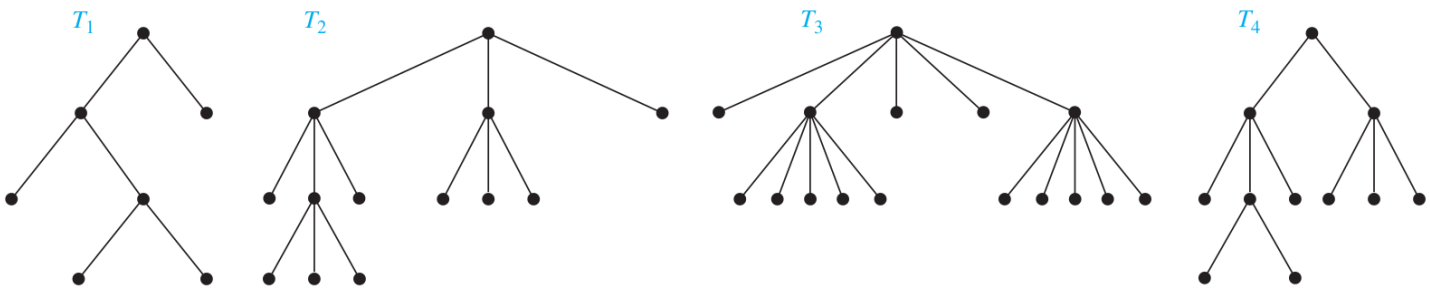


Figure: Rosen, p. 748

- T_1 is a full binary tree, T_2 is a full 3-ary tree, T_3 is a full 5-ary tree.
- T_4 not full (why?)

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Some tree models

Example: Family tree.

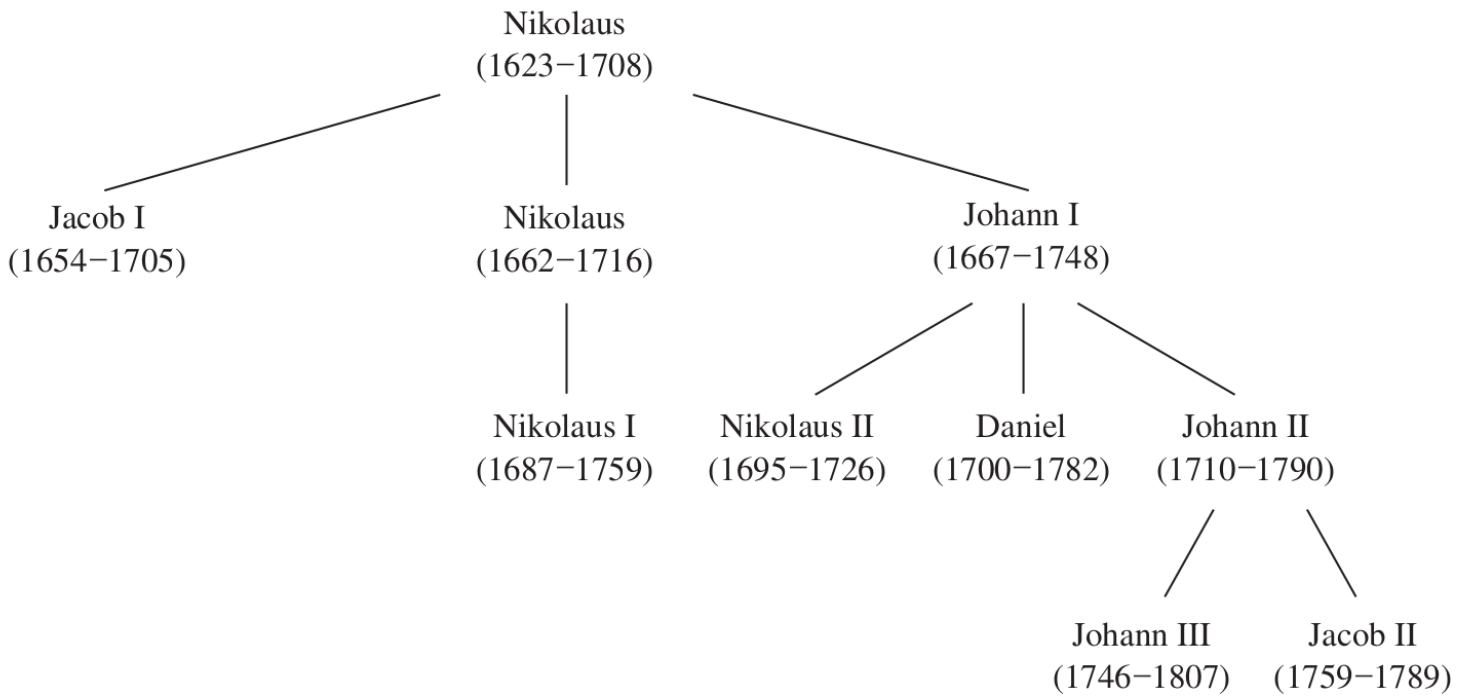


Figure: Rosen, p. 745

Some tree models

Example: Organization.

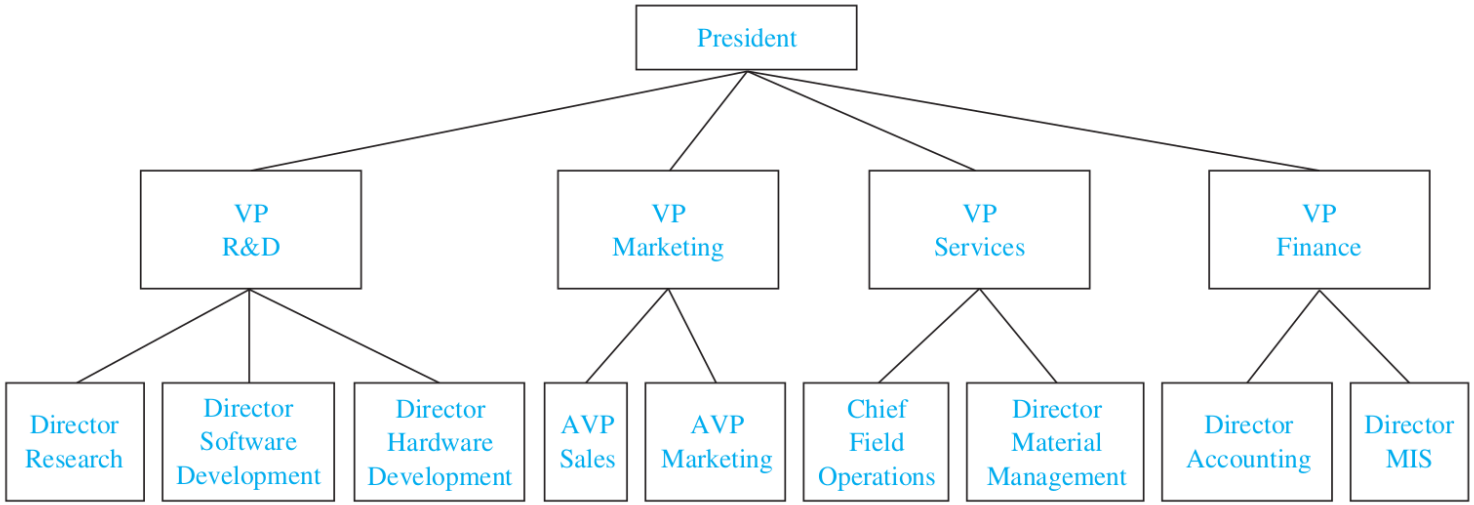


Figure: Rosen, p. 751

Some tree models

Example: Computer file system.

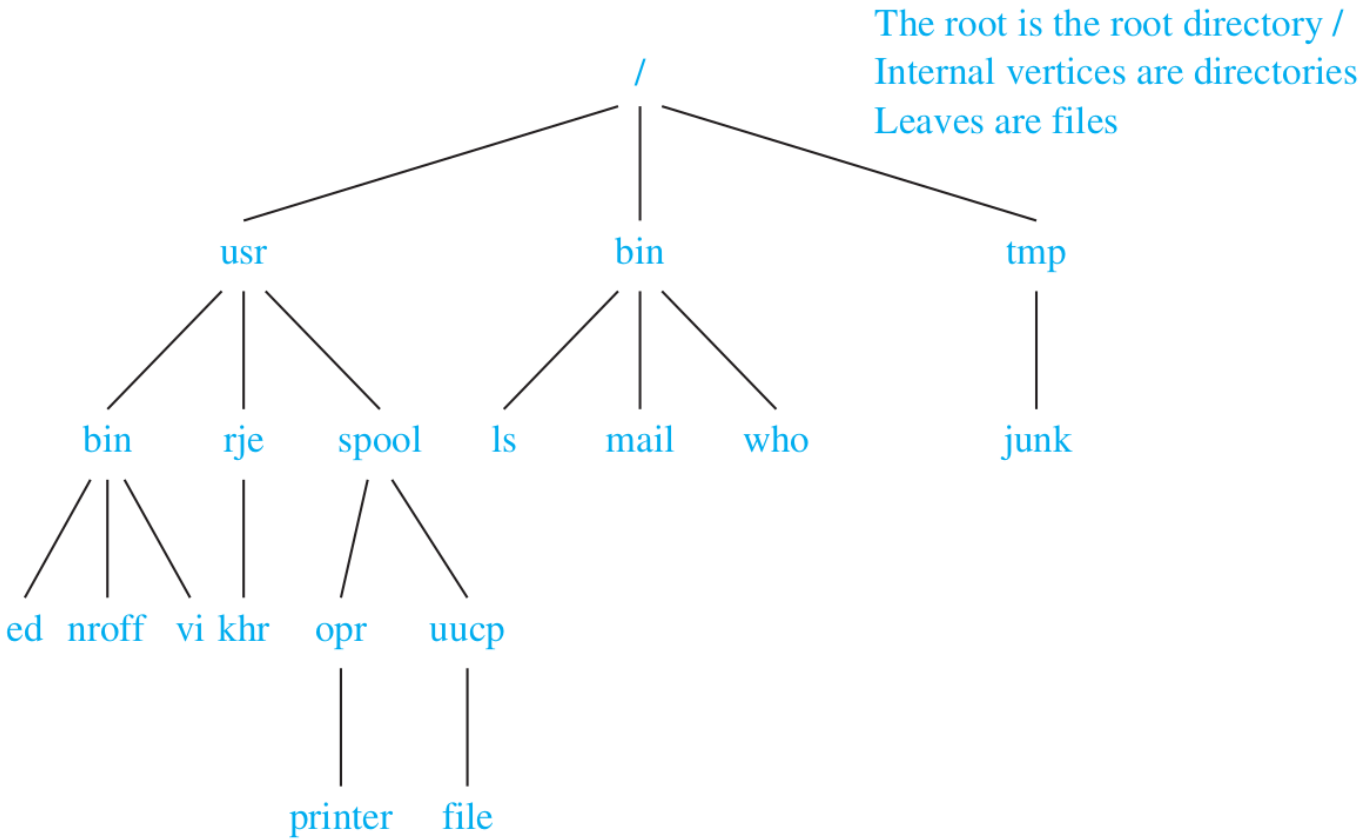


Figure: Rosen, p. 751

Some tree models

Example: Arithmetic expression.

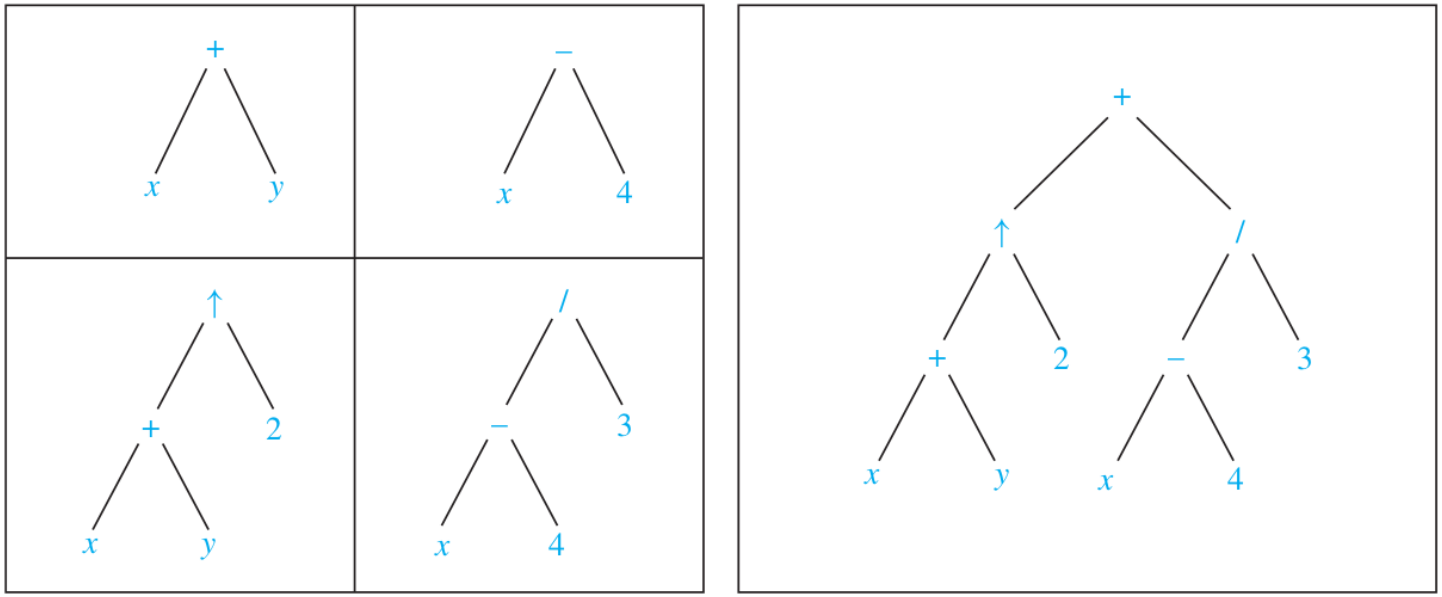


Figure: Rosen, p. 780

Fundamental properties of trees

Theorem

Let G be a simple graph with n vertices. The following statements are equivalent:

1. *G is a tree.*
2. *G is connected and has no circuits.*
3. *G is connected and has $n-1$ edge.*
4. *G has no circuits and has $n-1$ edge.*
5. *G is connected but the removal of any edge disconnects it (in other words, every edge is a bridge).*
6. *There is a unique path between any pair of vertices.*
7. *G has no circuit, but if we add a new edge connecting two non-adjacent vertices of G, then the new graph has exactly one circuit.*

Some properties of full m-ary trees

Theorem

Let n, i, ℓ be the numbers of vertices, internal vertices and leaves of a full m-ary tree, respectively. These three quantities are related to each other by the following equations:

1. $$i = \frac{n-1}{m}, \ell = \frac{(m-1)n+1}{m}.$$
2. $$n = mi + 1, \ell = (m-1)i + 1.$$
3. $$n = \frac{m\ell - 1}{m-1}, i = \frac{\ell - 1}{m-1}$$